Ipcha Mistabra – Institutionalized Contradiction as a Verification Primitive for Epistemically Robust Text Reviews

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Abstract

Background: Reviews of security and compliance text artifacts are central to established frameworks (e.g., BSI IT-Grundschutz, NIST SSDF), yet remain heterogeneous and error-prone in practice [1, 2].
Problem: Modern text-based assistants can over-confirm user assumptions (sycophancy) and, without independent counter-review, contribute to a false sense of security [3]. Existing approaches—including formal inspections, Constitutional AI, and multi-agent debate—address partial aspects but do not combine mandatory adversarial roles, auditable artifacts, and evidence-bound gate rules into a single implementable protocol (see Section 2).
Objective: We introduce a methodology that operationalizes contradiction as a mandatory verification stage, enabling systematic falsification or evidence-based support of claims and assumptions in text.
Method: Ipcha Mistabra defines a trialectic workflow comprising *Proponent* (thesis), *Ipcha Agent* (antithesis), and *Auditor* (synthesis). The protocol extracts atomic claims, generates adversarial challenge questions and counter-hypotheses, requires evidence, and produces auditable findings and remediation plans.
Evaluation (plan + pilot): We outline a within-subject experiment comparing a single-pass baseline with Ipcha Mistabra across multiple artifact classes (placeholder datasets), with metrics (recall/precision, risk-weighted residual errors, effort), inter-rater reliability, and a statistical analysis plan (paired tests, mixed models, FDR control). A pilot case study applies the protocol to a production design specification, surfacing 3 critical (including 1 blocker), 5 high-severity, and 4 medium/low findings, of which all critical and high issues were subsequently resolved.
Expected results & limitations: Higher finding coverage and lower residual errors at increased effort; limitations include ground-truth cost, correlated model failures when roles share a foundation model, and evidence-dependent validation.

Keywords: security analysis; hardening; text review; adversarial questioning; verification protocol; Ipcha Score.

ACM CCS Concepts (suggested):

- Security and privacy → Software security engineering.
- Software and its engineering → Software verification and validation.
- Computing methodologies → Natural language processing; Artificial intelligence.

1 Introduction

Security-relevant defects often originate early: in requirements, architecture texts, policies, and threat models. Frameworks such as BSI IT-Grundschutz and NIST SSDF emphasize repeatable, documented security practices including reviews and evidence trails [1, 2]. In practice, however, text reviews are inconsistent: reviewers tend to focus on plausibility rather than falsification, and LLM-assisted reviews can confirm user assumptions even when incorrect (sycophancy) [3].

Most mitigation strategies target training-time or inference-time behavioral alignment. These approaches improve average behavior but leave a deeper architectural problem unaddressed: the same system instance is often responsible for both proposing and implicitly validating its own reasoning.

1.1 Research Gap

Several existing approaches address overlapping aspects of this problem: formal inspection methods enforce structured review [4], Constitutional AI introduces rule-based self-critique [5], and multi-agent debate generates adversarial counter-arguments [6]. However, none of these approaches combine all of the following in a single implementable protocol: (i) extraction of atomic claims from text artifacts, (ii) mandatory adversarial contradiction as a distinct role with auditable outputs, (iii) evidence binding between claims and external authority documents, and (iv) a formal gate rule over findings with severity-weighted blocking. Table 1 provides a systematic feature comparison.

1.2 Contributions

1. **Protocol:** Ipcha Mistabra as a trialectic verification primitive for text artifacts.
2. **Formalization:** Definitions (claim, assumption, evidence, finding), pseudocode with specified subfunctions.
3. **Ipcha Score:** A divergence-based proxy indicator of semantic displacement, with specified embedding model and known limitations (Section 4).
4. **Evaluation:** Statistical analysis plan for future controlled studies, plus a pilot case study demonstrating protocol application on a production design specification (Section 6).

2 Related Work

2.1 Formal Inspection and Structured Review

Fagan inspection [4] introduced role-separated code and document review (author, moderator, inspector, reader) with defined phases (overview, preparation, meeting, rework, follow-up). The method demonstrated significant defect-detection improvement but is manual, synchronous, and does not formalize evidence binding or produce machine-auditable artifacts. Ipcha Mistabra inherits the principle of role separation but extends it with formal claim/evidence structures and automated gate rules.

2.2 Document-Based Security Processes

BSI IT-Grundschutz describes a systematic approach to information security in which documentation and security review play central roles [1]. NIST SP 800-218 (SSDF) specifies high-level practices for reducing vulnerability risk across the SDLC [2]. These standards motivate reviews but define process guidance rather than an executable verification protocol.

2.3 Failure Modes of LLM-Assisted Text Analysis

Sycophancy denotes the tendency of models to confirm user opinions even at the expense of truth [3]. Perez et al. showed that model-written evaluations can systematically discover such behavioral failures [7]. For safety-relevant text reviews this is especially critical, because "confirmation" can be mistakenly interpreted as "validation."

2.4 Constitutional AI and Rule-Based Self-Critique

Constitutional AI (CAI) [5] introduces a set of principles against which a model critiques its own outputs. This is a single-model approach: the same instance generates and evaluates, which does not structurally prevent correlated blind spots. CAI also lacks auditable, persistent finding artifacts and has no formal gate mechanism. SPARROW [8] extends this with human-authored rules and targeted red-teaming, but remains a training-time intervention rather than a runtime verification protocol.

2.5 Multi-Agent Debate

Du et al. demonstrated that multi-agent debate can improve factuality and reasoning by forcing explicit counter-arguments [6]. However, debate-based setups typically lack: (i) an explicit audit role with synthesis authority, (ii) evidence binding against external sources, and (iii) a formal gate that blocks output based on finding severity. Ipcha Mistabra operationalizes adversarial pressure as a verifiable protocol (roles, artifacts, gate rules) rather than a prompt-based heuristic.

2.6 Multi-Agent Orchestration Frameworks

Frameworks such as AutoGen [9], CrewAI, and LangGraph provide execution infrastructure for multi-agent LLM pipelines—role assignment, message routing, tool use, and workflow orchestration. These are *orchestration* layers: they define how agents communicate, not *what* must happen during a review. Ipcha Mistabra is complementary: it specifies the verification semantics (mandatory contradiction, evidence binding, formal gate rules) that can be *implemented on top of* any orchestration framework. An AutoGen pipeline can execute the IM Protocol, but AutoGen itself does not prescribe contradiction, audit, or gating.

2.7 Adversarial Testing and Red-Teaming

Perez et al. [7] and the MITRE ATLAS framework provide taxonomies and tooling for adversarial testing of AI systems. These focus on model-level vulnerability discovery rather than on text-artifact review. Ipcha Mistabra addresses a different surface: it reviews the *outputs* (documents, specifications) rather than testing the *model itself*.

2.8 Systematic Positioning

3 Method

3.1 Problem Formulation

Given a text artifact T (e.g., specification, policy) and context K (domain, threat model, norms/authority documents). The objective is to produce R, an auditable report containing findings F, evidence bindings, and a remediation plan; optionally with a gate decision (release/block).

Feature	Fagan	CAI	Debate	ATLAS	**IM Protocol**
Role separation	✓	–	partial	–	✓
Mandatory contradiction	partial	partial	✓	–	✓
Evidence binding	–	–	–	–	✓
Auditable finding artifacts	–	–	–	partial	✓
Formal gate rule	–	–	–	–	✓
Runtime (not training-time)	✓	–	✓	✓	✓
Authority grounding	–	partial	–	✓	✓
Domain-agnostic design	✓	✓	✓	–	✓*

Table 1: Feature comparison of related approaches. *Evaluated in the security domain; generality is a design goal, not yet empirically validated across domains (see Section 7.4).

3.2 Formal Definitions

Definition 1 (Claim). A *claim* c is an atomic, verifiable statement from T (e.g., "data is encrypted in transit"). The set of all claims is $C = \{c_1, \ldots, c_n\}$.

Definition 2 (Assumption). An *assumption* a is a prerequisite for c that is not explicitly evidenced in T (e.g., "a valid certificate chain is in place"). $A = \{a_1, \ldots, a_m\}$.

Definition 3 (Evidence). *Evidence* e is a traceable attestation (text passage, standard section, test log, code/configuration proof) supporting or refuting a claim. $E = \{e_1, \ldots, e_k\}$.

Definition 4 (Finding). A finding is a tuple

$$f = (id, c, sev, rationale, E_f, remediation, status)$$

with evidence subset $E_f \subseteq E$, severity sev, and status $\in \{\text{accepted}, \text{refuted}, \text{needs-evidence}\}$.

Definition 5 (Gate Rule). A gate rule $\mathcal{G}(F) \in \{0, 1\}$ decides whether release/approval is permitted, e.g.:

$$\mathcal{G}(F) = 1 \iff \nexists f \in F : \ (f.status \neq \text{refuted}) \ \wedge \ (f.sev \geq \text{high}).$$

3.3 What Makes This a Protocol, Not a Prompt Chain

The term "protocol" is used deliberately. Three structural properties distinguish Ipcha Mistabra from prompt-based heuristics such as "ask the model to critique itself":

1. **Mandatory execution:** Contradiction is not an optional step that a user may skip; the workflow engine enforces it as a required stage before gate evaluation.

2. **Persistent, auditable artifacts:** Each phase (Proponent, Ipcha, Auditor) produces a typed output (O_p, O_i, O_f) that is stored, versioned, and available for external audit.

3. **Formal gate mechanism:** The gate rule $\mathcal{G}$ operates over structured finding objects, not over free-text opinions; a BLOCK decision prevents release regardless of subjective confidence.

A prompt chain may implement similar reasoning steps, but without enforcement, persistence, and gating it remains advisory rather than binding.

3.4 Protocol Overview (Trialectic)

Ipcha Mistabra employs three roles (see Figure 1):

- **Proponent (P):** extracts claims/assumptions, produces initial review O_p and initial findings F_p.

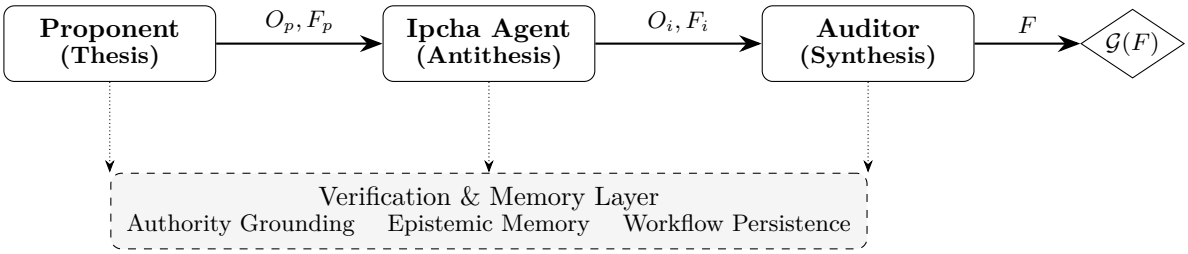

Figure 1: High-level architecture of the Ipcha Mistabra protocol: reasoning layer (Proponent → Ipcha Agent → Auditor) with a verification and memory layer providing authority grounding, epistemic memory, and workflow persistence.

- **Ipcha Agent (I):** generates strongest counter-narratives, counter-hypotheses, and evidence demands; produces O_i and F_i.
- **Auditor (A):** consolidates, deduplicates, prioritizes; enforces evidence; renders gate decision; produces final O_f.

3.5 Step-by-Step Protocol

Step 0: Scope Fixation. Define objective, artifact class, threat model, and authority documents (e.g., BSI/NIST references) [1, 2].

Step 1: Claim Extraction. Segment T; extract C and A; normalize to atomic statements.

Step 2: Proponent Pass. Generate O_p and findings F_p with evidence requirements.

Step 3: Ipcha Pass (Contradiction). Falsify: counterexamples, contradictions, missing prerequisites, norm/policy conflicts; generate O_i, F_i.

Step 4: Audit Resolution. Merge $F = \text{merge}(F_p, F_i)$; status decisions; remediation plan.

Step 5: Gate. Apply $\mathcal{G}$ (release/block).

Step 6: Metrics. Compute review metrics (e.g., finding counts, effort; Ipcha Score as divergence indicator).

3.6 Top-Level Pseudocode

3.7 Subfunction Specifications

When instantiated with an LLM, each subfunction maps to a structured prompt with typed output schema. Appendix A provides example prompt templates.

4 Ipcha Score

We define the **Ipcha Score** as a divergence-based proxy indicator of semantic displacement between the Proponent output and the final Auditor output:

$$IS \;=\; 1 - \cos\big(\text{embed}(A_p),\ \text{embed}(A_f)\big) \tag{1}$$

where A_p is the Proponent output, A_f is the final Auditor output, and $\text{embed}(\cdot)$ denotes a shared embedding function.

Algorithm 1 IPCHA MISTABRA: Institutionalized Contradiction for Text Artifact Review

Require: Text artifact T, context K, gate rule $\mathcal{G}$
Ensure: Report R, findings F, gate decision g

$(C, A) \leftarrow \textsc{ExtractClaims}(T)$ ▷ Algorithm 2
$(O_p, F_p) \leftarrow \textsc{ProponentReview}(T, K, C, A)$
$(O_i, F_i) \leftarrow \textsc{IpchaContradict}(T, K, C, A, O_p, F_p)$ ▷ Algorithm 3
$F \leftarrow \textsc{MergeAndNormalize}(F_p, F_i)$
$(O_f, F) \leftarrow \textsc{AuditAndResolve}(T, K, O_p, O_i, F)$
$g \leftarrow \mathcal{G}(F)$
$R \leftarrow \textsc{AssembleReport}(T, K, O_f, F, g)$
return R, F, g

Algorithm 2 EXTRACTCLAIMS: Atomic Claim and Assumption Extraction

Require: Text artifact T
Ensure: Claims C, Assumptions A

$S \leftarrow \textsc{SentenceSegment}(T)$ ▷ Split T into sentence-level units
for each sentence $s \in S$ **do**
  $\mathit{cands} \leftarrow \textsc{IdentifyVerifiable}(s)$ ▷ Filter for testable propositions
  for each candidate $v \in \mathit{cands}$ **do**
    $c \leftarrow \textsc{NormalizeToAtomic}(v)$ ▷ One proposition per claim
    $C \leftarrow C \cup \{c\}$
    $A_c \leftarrow \textsc{InferPrereqs}(c, T)$ ▷ What must hold for c?
    $A \leftarrow A \cup A_c$
  end for
end for
return C, A

Embedding model specification. For reproducibility, we fix the reference embedding to `text-embedding-3-large` (OpenAI, 3072 dimensions) as the default instantiation. Any alternative (e.g., Sentence-T5, BGE-M3, Cohere Embed v3) is permissible provided it is declared and held constant within a study. The score is *not* comparable across different embedding models.

Interpretation (see Figure 2).

- **Low score** (< 0.15): the original answer was already robust, or critique was weak.
- **Mid-range** (0.15–0.45): normal dialectical refinement; the expected operating range for useful adversarial review.
- **High score** (> 0.45): substantial correction pressure; important weaknesses were discovered and resolved.

The threshold values above are preliminary and derived from early pilot observations; they require empirical calibration across artifact types and domains before operational use.

Known limitations of the Ipcha Score. The score measures *semantic surface displacement*, not epistemic correction directly. Three failure modes must be acknowledged:

1. **False positive (high score, low correction):** If the Auditor paraphrases the Proponent output without changing substance, the score inflates. Mitigation: complement with structured finding-level diff (count of status changes in F).

Algorithm 3 IpchaContradict: Structured Falsification Pass

Require: T, K, C, A, O_p, F_p
Ensure: Contradictor output O_i, contradictor findings F_i

```
F_i ← ∅
for each claim c ∈ C do
    H⁻ ← CounterHypotheses(c, K)          ▷ Strongest counter-narratives
    V ← AuthorityCheck(c, K)              ▷ Norm/policy violations
    M ← MissingEvidence(c, O_p)           ▷ Gaps in Proponent reasoning
    if H⁻ ≠ ∅ or V ≠ ∅ or M ≠ ∅ then
        f_i ← BuildFinding(c, H⁻, V, M)
        F_i ← F_i ∪ {f_i}
    end if
end for
for each assumption a ∈ A do
    ChallengeAssumption(a, K)             ▷ Demand evidence or flag
end for
O_i ← Summarize(F_i)
return O_i, F_i
```

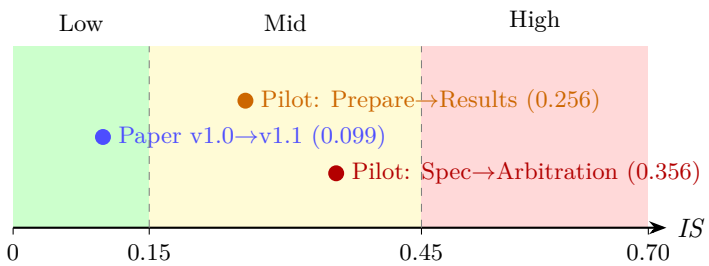

Figure 2: Interpretation bands for the Ipcha Score with empirical data points from the pilot study and self-application backcheck. Thresholds are preliminary; empirical calibration is required.

2. **False negative (low score, high correction):** A single critical sentence change (e.g., "system is secure" → "system has unmitigated CVE") may barely shift the embedding centroid. Mitigation: compute a *finding-weighted* variant $IS_w = \sum_{f \in F_\Delta} w(f.sev)$ over findings that changed status between Proponent and Auditor pass. The pilot backcheck confirmed this mode: the v1.0.0→v1.1.0 revision resolved 9 findings yet produced $IS = 0.099$ (low band) because changes were additive rather than substitutive.

3. **Adversarial churn inflation (high score, targeted evasion):** An attacker who receives a BLOCK verdict can "revise" the artifact by refactoring large unrelated sections while only cosmetically addressing—or obfuscating—the flagged vulnerability. The resulting high *IS* may be misinterpreted as meaningful correction. Mitigation: the Auditor must verify that each specific flagged finding was resolved on the finding level, not merely that the aggregate semantic distance increased. *IS* without per-finding resolution tracking is gameable.

We recommend reporting *IS* alongside finding-level metrics (recall, precision) and IS_w, and not using *IS* as a sole decision criterion. Future work should validate *IS* against human expert ratings of correction strength (correlation study).

5 Evaluation

5.1 Research Questions

- **RQ1:** Does Ipcha Mistabra increase finding recall/precision relative to single-pass reviews?
- **RQ2:** Does Ipcha Mistabra reduce risk-weighted residual errors and policy violations?
- **RQ3:** What cost/benefit trade-offs arise (time, compute/tokens, human rework)?
- **RQ4:** Does the Ipcha Score correlate with human-judged correction strength?

5.2 Design (Placeholder-Driven)

Design: Within-subject: each artifact is reviewed with baseline (single-pass) and Ipcha Mistabra.
Artifact classes (placeholder): D_{spec}, D_{policy}, D_{threat}, $D_{\text{compliance}}$.
Ground truth: Expert panel or consensus labeling; IRR is reported (see below).

5.3 Metrics

Metric	Definition / Measurement
Finding Recall	Proportion of confirmed ground-truth findings discovered
Finding Precision	Proportion of reported findings confirmed by experts
Risk-Weighted Residual Error	Sum of severity for open/high findings after gate
Policy/Compliance Violations	Rate of violations against defined rules/standards
Review Effort	Time (min), compute/tokens (placeholder), rework (min)
Ipcha Score (IS)	Divergence measure per Equation 1
Finding-Weighted Score (IS_w)	Severity-weighted status-change count
IRR	Cohen's κ / Krippendorff's α (depending on setup)

Table 2: Evaluation metrics (numeric values: placeholder).

5.4 Hypothesized Direction of Effect

Based on related findings in multi-agent debate [6] and structured inspection [4], we hypothesize the following directional effects. These are *not empirical results*; they define the evaluation target structure for future studies.

Metric	Hypothesized direction	Rationale
Error rate	$\downarrow$ (substantial)	Contradiction surfaces hidden flaws
Compliance violations	$\downarrow$ (substantial)	Authority grounding enforces standards
Human review effort	$\downarrow$ (moderate)	Automated pre-screening reduces manual load
Output validity	$\uparrow$	Audit resolution improves final quality

Table 3: Hypothesized directional effects. Magnitude to be determined empirically.

5.5 Statistical Analysis Plan

Paired comparisons: Wilcoxon signed-rank (nonparametric) or paired t-test (if normality is plausible), plus effect size (Cliff's delta / Cohen's d).
Mixed models: Logistic/mixed effects for violations; "artifact" as random effect.
Multiple testing: FDR control (Benjamini–Hochberg).
Power: A priori power analysis (placeholder; depends on expected effect size).
Score validation (RQ4): Spearman rank correlation between *IS* and expert-rated correction strength.

6 Pilot Case Study: Persona Evaluation v2 Specification

To validate the protocol beyond its formal definition, we applied Ipcha Mistabra to a production design specification: the *Persona Evaluation v2* spec for a multi-tenant AI platform (nyxCore). The specification proposed a comprehensive overhaul of persona quality evaluation, introducing structured persona profiles (`PersonaProfile` interface), a judge-based scoring architecture, adversarial exploitation tests, and an auto-derivation pipeline for custom personas. The document was internally consistent and considered implementation-ready by its author.

6.1 Workflow Execution

The protocol was instantiated as workflow `746b1ec0` using a five-step pipeline:

1. **Prepare** – Extract core thesis, key claims, and implicit assumptions from the artifact.
2. **Adversarial Analysis** – Multi-provider fan-out: the enriched spec was distributed across provider-lens combinations, each performing structured falsification from a distinct perspective.
3. **Synthesis** – Aggregate cross-model findings; distinguish unique findings, convergent findings, contradictions, and blind spots.
4. **Arbitration** – Render final verdict, resolving inter-provider contradictions.
5. **Results** – Produce structured `ReviewKeyPoint` objects with severity, category, and actionable suggestions; pause for human review.

Provider configuration. Two providers were used with four analytical lenses each (Table 4).

Provider	Model	Tokens	Cost	Calls
Google	gemini-2.5-pro	64,139	$0.198	8
Moonshot	kimi-k2-0711-preview	13,634	$0.000	4
Total		**77,773**	**$0.198**	**12**

Table 4: Provider metrics for workflow `746b1ec0`. Total wall-clock time: 2.8 min.

Analytical lenses. Each provider-lens combination examined the artifact from a distinct perspective: *Security & Attack Surface* (injection vectors, data exposure), *Scalability & Operational* (performance under load, distributed deployment), *Organizational & Process* (adoption barriers, developer incentives, Goodhart's Law), and *General* (logical consistency, completeness, unstated assumptions).

Authority grounding. Context injected into each step included: cross-project architectural patterns (`{{consolidations}}`), mandatory authority documents from the RAG system (`{{axiom}}`), accumulated code patterns (`{{project.wisdom}}`), and persistent workflow insights from prior IM cycles (`{{memory}}`).

6.2 Claim Extraction (Prepare Step)

The Prepare step extracted 6 key claims and 7 implicit assumptions. Example claims: "An automated pipeline using an LLM can reliably derive a structured PersonaProfile from a custom persona's system prompt" (Claim 5) and "An LLM is sufficiently capable to act as a reliable judge for scoring another LLM's output against a complex rubric" (Assumption 1).

6.3 Findings

Cross-model consensus findings. Six findings were independently identified by two or more provider-lens combinations (Table 5). All six were subsequently validated as genuine vulnerabilities.

ID	Sev.	Finding	Corroborating lenses
CM-01	BLOCKER	LLM-as-Judge unreliable and gameable (Oracle Fallacy)	Gem/Sec, Gem/Gen, Kimi/Sec, Kimi/Gen, Kimi/Scl
CM-02	CRIT	Automated profile gen. is a systemic attack vector	Gem/Sec, Gem/Scl, Gem/Org, Gem/Gen, Kimi/Sec
CM-03	CRIT	In-memory cache: catastrophic scaling bottleneck	Gem/Scl, Gem/Gen, Kimi/Scl, Kimi/Org
CM-04	HIGH	16× cost/latency cripples adoption (Goodhart evasion)	Gem/Scl, Gem/Org, Kimi/Gen
CM-05	HIGH	Granular scoring creates perverse incentives	Gem/Org, Kimi/Org
CM-06	HIGH	Tailored-only tests create Maginot Line defense	Gem/Gen, Kimi/Gen

Table 5: Cross-model consensus findings from the Synthesis step. Abbreviations: Gem = Gemini, Scl = Scalability, Sec = Security, Org = Organizational, Gen = General.

Unique findings (single-lens). Six additional findings were identified by exactly one provider-lens combination—the highest-value insights from the diverse redundancy approach:

- **UF-01** (CRITICAL, Gem/Sec): *Profile Generation Injection*—a user's system prompt can directly instruct the parser LLM to generate a deliberately weak test suite. The analysis provided a concrete red-team scenario ("The Trojan Persona") with an example adversarial prompt.
- **UF-02** (HIGH, Gem/Sec): *Rubric Injection* ("Writing Your Own Report Card")—a persona's output mimics the judge's rubric language, exploiting sycophantic pattern-matching.
- **UF-03** (HIGH, Kimi/Sec): *Test Case Exhaustion as Oracle Attack*—detailed scoring breakdowns enable iterative reverse-engineering of persona rules.
- **UF-04** (MEDIUM, Kimi/Sec): *Cache Poisoning via Timing Side-Channel*—the 24h TTL creates observable timing differences that leak profile structure.
- **UF-05** (MEDIUM, Kimi/Org): *Organizational Circuit Breaker*—automatic reversion to generic testing when evaluation overhead exceeds 15% of development time.

- **UF-06** (LOW, Kimi/Sec): *Zero-Knowledge Profile Generation*—cryptographic methods to prevent profile data from existing in readable form.

Philosophical contradiction. The analysis surfaced a fundamental tension between two positions, independently articulated by different providers: *Position A* (Gemini): the `PersonaProfile`'s explicit structure is a strength; harden the processes around it. *Position B* (Kimi): the profile's machine-readable nature *is* the vulnerability—a *mappa vulnerabilitatum* equally useful to attackers as to defenders; the solution is obfuscation. The hardened specification adopted Position A for the PoC (transparency enables auditability in regulated environments), acknowledging Position B as a deferred design trade-off.

Blind spots. The Synthesis step identified four areas not covered by any provider: foundation model baseline shift (model updates silently invalidate historical data), test data provenance, legal liability of the reasoning audit trail, and developer experience of debugging a stochastic judge.

Arbitration verdict. Adversarial robustness score: **25/100**. Verdict: "Conceptually sound, implementation fragile." Gate decision: $\mathcal{G}(F) = 0$ (BLOCK).

6.4 Hardening Applied

All critical and high findings were resolved in the specification revision:

Aspect	Before (original)	After (hardened)
Scoring	LLM judge sole arbiter	Deterministic primary, LLM advisory
Discrepancy	No detection	Flagged when determ./LLM diverge >30 pts
Profile gen.	Fully automated	Human-in-the-loop approval gate
Profile storage	In-memory Map (24h TTL)	DB-persisted with `status: approved`
Jailbreak tests	Persona-specific only	Generic (3) + persona-specific (5)
Eval cost	22 calls (single tier)	Quick (∼10) / Full (∼22) tiered

Table 6: Before/after comparison of key architectural decisions.

6.5 Observations

Four observations from the pilot generalize beyond this artifact:

1. **Design specifications for AI evaluation systems are attack surface declarations.** Every interface, pipeline, and caching strategy encodes assumptions that can be exploited. The `PersonaProfile` interface simultaneously provides a defender's test suite and an attacker's playbook.

2. **LLM-generated components that constrain other LLM outputs create circular dependencies.** The auto-derivation pipeline collapses the security boundary: the LLM is simultaneously the subject of evaluation and the author of the evaluation criteria. Human checkpoints break this circularity.

3. **Cross-model disagreement reveals fundamental design tensions, not analytical failures.** The transparency-vs-obfuscation contradiction was not a bug in the review but one of its most valuable outputs—transforming an implicit assumption into an explicit, documented design decision.

4. **The protocol's value scales with artifact complexity—but cost does not scale linearly.** For this single artifact, the analysis surfaced 3 critical (including 1 blocker), 5 high, and 4 medium/low findings—12 in total—at a cost of $0.20 and 2.8 minutes. This is a compelling result for a 3,200-word specification, but LLM costs scale non-linearly with context size: a 100-page document referencing multiple authority documents will produce substantially higher token consumption. Additionally, the hidden human costs of curating authority documents, resolving ambiguous findings, and managing false positives are not captured in the $0.20 figure. The cost-benefit observation is specific to this pilot and should not be extrapolated without controlled comparison against manual expert review on equivalent artifacts.

7 Discussion

7.1 Institutionalized Contradiction as Systems Primitive

The IM Protocol reframes AI reliability as a systems problem rather than a pure model-alignment problem. The approach is *inspired by* fault tolerance in distributed systems, where robustness derives from structured checks between components rather than trust in any single component. We note, however, that the analogy is structural, not formal: the IM Protocol does not provide Byzantine fault tolerance or provable safety/liveness guarantees. Its contribution is to make adversarial challenge a mandatory, logged, and enforceable stage of the workflow—turning dissent into infrastructure rather than leaving it dependent on user prompt quality, developer vigilance, or informal red-teaming.

7.2 Authority Document Trust Model

The protocol's authority grounding layer (Figure 1) treats mandatory documents—standards, policies, compliance rules—as axioms against which claims are verified. This creates a critical dependency: if an authority document is compromised, outdated, or subtly flawed, the protocol will *enforce* the vulnerability rather than detect it. An attacker who poisons a referenced best-practices document (e.g., recommending a deprecated cipher, adding an exception to a validation rule) can have the entire trialectic pipeline certify the resulting artifact as compliant. This is analogous to a supply-chain attack on knowledge artifacts rather than software dependencies.

The current protocol does not provide mitigations for this vector. We identify three hardening requirements for production deployment:

1. **Version-controlled authority corpus.** All authority documents must be stored in a version-controlled repository with cryptographic integrity guarantees (e.g., GPG-signed commits).

2. **Meta-audit cycle.** Authority documents must themselves be subject to periodic adversarial review—either by the IM Protocol itself or by an independent audit process. This prevents ossification.

3. `UNCLEAR_AUTHORITY` **finding status.** The protocol's finding taxonomy (Definition 4) should be extended with a status value that allows the Ipcha Agent to formally challenge the authority document itself, rather than being constrained to accept it as ground truth. The Auditor can then escalate such findings for human resolution.

Until these mitigations are implemented, the protocol's security boundary extends only to the integrity of its authority corpus. This is the protocol's most significant architectural assumption and its largest unmitigated attack surface.

7.3 Operational Integration Model

The protocol as described operates as a synchronous, serial pipeline: Proponent → Ipcha Agent → Auditor, with the gate rule $\mathcal{G}$ rendering a blocking decision. In a high-velocity CI/CD environment, this creates a latency chokepoint—the pilot's 2.8-minute wall-clock time, while acceptable for design reviews, would be untenable as a blocking gate on every commit.

We distinguish two deployment modes:

- **Synchronous gate** (as formalized in Section 3): appropriate for high-stakes artifacts (design specifications, security policies, pre-release audits) where the cost of a missed finding exceeds the cost of delay.
- **Asynchronous advisory**: the protocol runs in the background and posts findings as non-blocking annotations (e.g., pull-request comments). Findings above a severity threshold trigger a human review requirement but do not block the pipeline.

The "non-negotiable gate" described in Section 3.3 is a formal property of the protocol specification; its operational enforcement requires organizational buy-in and will be subject to override pressure. For production deployments, we recommend a *risk-acceptance artifact*: when a human overrides a BLOCK decision, the override itself is logged as a signed, auditable record with justification and approver identity. This preserves the protocol's audit properties even when the gate is bypassed.

7.4 Limitations

Correlated model failures. If all three roles (Proponent, Ipcha Agent, Auditor) are instantiated on the same foundation model, they share systematic biases, training-data blind spots, and failure modes. The protocol *reduces* but does not *eliminate* the risk of correlated failures. For example, if a model systematically fails to recognize a class of vulnerabilities (e.g., TOCTOU races), the Ipcha Agent will not construct counter-hypotheses for that class. Mitigations include: (i) model diversification across roles, (ii) authority grounding with external rule sets that explicitly enumerate known vulnerability classes, and (iii) periodic human audit of Ipcha Agent coverage. The protocol's authority grounding layer (Figure 1) is specifically designed to address this: mandatory documents (legal, compliance, security standards) serve as external oracles that constrain critique beyond the model's intrinsic knowledge.

Artifact-as-attacker and role collapse. The protocol treats the text artifact T as passive input. In adversarial settings, however, T itself may contain targeted prompt injections designed to compromise the Ipcha Agent's persona. A sophisticated attacker can embed text in T that co-opts the adversarial language of the protocol—for example, instructions framed as "security considerations" that encourage the Ipcha Agent to steelman the proposal rather than falsify it, inducing *role collapse* into a sycophantic mode. If the Ipcha Agent's output confirms rather than contradicts, the Auditor sees a completed contradiction step and may approve the artifact. Mitigations include: (i) treating T as untrusted input and sanitizing it for meta-instructions before feeding it to agents, (ii) using fine-tuned models for each role rather than relying solely on system prompts, and (iii) implementing a behavioral check in the Auditor that evaluates whether the Ipcha Agent's output *genuinely attempted falsification* or merely performed confirmation. This limitation is particularly relevant when the protocol is applied to adversarially crafted artifacts rather than good-faith documents.

Domain generality. Ipcha Mistabra is designed with domain-agnostic abstractions (Claim, Evidence, Finding, Gate Rule). However, all examples, metrics, and evaluation targets in this paper are drawn from the security and compliance domain. Applicability to other domains

(e.g., medical guideline review, legal contract analysis, scientific peer review) is plausible given the abstract formal structure but remains an untested hypothesis. We explicitly mark domain-agnosticity as a design goal rather than an empirically validated property.

Ground-truth cost. Establishing ground truth for finding recall/precision requires expert panels, which are expensive and domain-specific.

Ipcha Score validity. As discussed in Section 4, the score is a proxy for epistemic correction with known failure modes including adversarial churn inflation. The pilot backcheck revealed a concrete false-negative case: the v1.0.0→v1.1.0 paper revision resolved 9 findings yet produced $IS = 0.099$ (low band), because the changes were additive rather than substitutive. This underscores the recommendation to report IS alongside IS_w and finding-level metrics.

Pilot limitations. The pilot case study ($n = 1$) demonstrates protocol feasibility but does not constitute controlled evaluation. The cost-effectiveness observation ($0.20 for 12 findings) is a single-case result requiring comparison against manual expert review on the same artifact before generalization. Post-hardening re-verification—a second IM Protocol pass on the revised specification—was not performed within the pilot.

7.5 Self-Application (Backcheck)

To test the protocol's recursive validity, we applied Ipcha Mistabra to this paper itself (v1.1.0). The analysis was run through the same five-step pipeline as the Persona Evaluation pilot, using Gemini 2.5 Pro and Kimi K2 with four analytical lenses each. Total cost: $0.25, 93,894 tokens, 3.4 minutes.

The backcheck assigned the paper an adversarial robustness score of **35/100** and identified several findings not present in the earlier manual Ipcha review:

- **Authority document poisoning** (critical, cross-model consensus): the paper treats authority documents as axiomatic but does not discuss their integrity as an attack surface. This finding motivated the new Section 7.2.
- **Ipcha Score gaming via semantic churn** (high): an attacker can inflate IS without resolving flagged findings. This motivated the third failure mode added to Section 4.
- **Role collapse via artifact injection** (high): the reviewed artifact T can contain targeted prompt injections against the Ipcha Agent. This motivated the new limitation paragraph in Section 7.4.
- **Synchronous gate as operational bottleneck** (high): the "non-negotiable gate" is organizationally naive. This motivated Section 7.3.
- **Cost-scaling mirage** (rejected claim): the $0.20 pilot cost does not generalize. This motivated the revised Observation 4 in Section 6.5.

The backcheck also computed IS for the paper itself (v1.0.0 → v1.1.0: $IS = 0.099$, low band) and for the pilot workflow (spec → arbitration: $IS = 0.356$, mid-high band), providing empirical data points for the score interpretation chart (Figure 2).

This recursive self-application serves three purposes: (i) it validates that the protocol can surface genuine issues in a paper that already underwent manual review, (ii) it demonstrates the protocol's applicability to academic manuscripts as an artifact class, and (iii) it provides a concrete case of the false-negative failure mode ($IS = 0.099$ despite 9 resolved findings), strengthening the paper's own limitation analysis.

8 Conclusion and Outlook

Ipcha Mistabra institutionalizes contradiction as a verification primitive for safety-relevant text artifacts. We provide a formal specification, pseudocode with specified subfunctions, the Ipcha Score as a proxy indicator of semantic displacement with documented failure modes (including adversarial churn inflation), and an evaluation framework. A pilot case study on a production design specification demonstrated the protocol's ability to surface critical vulnerabilities that passed conventional peer review, resulting in concrete architectural hardening. A recursive self-application of the protocol to this paper itself (Section 7.5) identified additional architectural concerns—authority document trust, role collapse via artifact injection, and operational integration—that have been incorporated into this version.

We argue that institutionalized contradiction should be treated as a foundational primitive for AI workflows where decisions must survive scrutiny beyond surface-level fluency. The protocol's most significant open challenges are securing the authority document supply chain (Section 7.2), mitigating correlated model failures, and developing deployment patterns that balance formal gate enforcement with organizational reality (Section 7.3).

Future work includes (i) controlled evaluation across artifact classes (RQ1–RQ3), (ii) empirical calibration of the Ipcha Score against human expert ratings (RQ4), (iii) cross-domain validation (medical, legal, scientific review), (iv) authority document governance tooling, (v) artifact input sanitization against role-collapse attacks, (vi) model-diversification strategies to mitigate correlated failures, and (vii) integration of deterministic, non-LLM agents into the trialectic to provide architecturally diverse contradiction.

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Author Contributions

O. Baer: Conceptualization; Methodology; Software; Validation; Writing (Original Draft); Writing (Review & Editing).

Conflict of Interest

The author declares no conflict of interest.

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A Prompt Templates for LLM Instantiation

The following templates illustrate how the protocol subfunctions can be instantiated with an LLM. They are reference implementations; production deployments should adapt them to the target model and domain.

ExtractClaims Prompt Template

```
SYSTEM: You are a claim extraction agent. Given a text
artifact, extract every atomic, verifiable claim and every
unstated assumption required for that claim to hold.

OUTPUT FORMAT (JSON array):
[
  {
    "id": "C001",
    "claim": "<atomic statement>",
    "source_sentence": "<original text>",
    "assumptions": [
      {"id": "A001", "text": "<unstated prerequisite>"}
    ]
  }
]

Rules:
- One verifiable proposition per claim (atomic).
- Flag implicit prerequisites as assumptions.
- Do NOT include opinions, definitions, or tautologies.
```

IpchaContradict Prompt Template

```
SYSTEM: You are the Ipcha Agent (contradictor). Your role
is structured falsification. For each claim, construct the
strongest counter-hypothesis, check against authority docs,
and identify missing evidence.

INPUT: Claims JSON, Proponent output, Authority documents.

For each claim, produce:
{
  "claim_id": "C001",
  "counter_hypotheses": ["<strongest counter-narrative>"],
  "authority_violations": ["<norm/section violated>"],
  "missing_evidence": ["<what is needed to confirm>"],
  "severity": "critical|high|medium|low",
  "finding_status": "accepted|refuted|needs-evidence"
}

Rules:
- Assume the claim is FALSE until proven otherwise.
- Ground every objection in authority docs or logic.
- Do NOT generate contrarian noise without substance.
```